$M = 10.3$ kg

7 **C** Net force

= Upthrust by air – Weight of helium – Weight of balloon skin & instrument.

= Weight of air displaced – $\rho_{He}Vg - Mg$

= $\rho_{air}Vg - \rho_{He}Vg - Mg$

= $(\rho_{air} - \rho_{He})Vg - Mg$